

MICROLAB[®] PSD/4



User Manual

Part Number 8892-01 (Rev. B)

HAMILTON
THE MEASURE OF EXCELLENCESM



Hamilton Company Instrument Warranty

Hamilton Company warrants this equipment (except valves*) to be free of defects in material and workmanship for 12 months from the date of receipt. This warranty is extended to the buyer of record on the original purchase order to Hamilton Company. Hamilton Company or an authorized Hamilton representative agrees to repair or replace, at its option and free of charge to the buyer at a normal place of business or at a Hamilton repair facility, any part or parts that under proper and normal use prove to be defective during the warranty period.** Abuse, unauthorized replacement of parts, modifications, or adjustments made by other than Company or its assigned representatives voids this warranty.

This warranty gives you specific rights. No other warranties, expressed or implied, including implications of warranties of merchantability and fitness for a particular product, are made. Hamilton Company's liability on the sale of all products shall be limited to repair, replacement, or refund of price of any defective product.**

Hamilton Company endeavors to provide prompt and satisfactory service.

*All Hamilton Company valves are warranted to be free of defects in material and workmanship at the time of delivery.

**Hamilton Company reserves the right to refuse to accept the return of any instrument or valve that has been used with radioactive or microbiological substances, or any other material that may be deemed hazardous to employees of Hamilton Company.

© June 2003 by Hamilton Company

GASTIGHT is a registered trademark of Hamilton Company.

MICROLAB is a registered trademark licensed to Hamilton Company.

TEFLON is a registered trademark of E. I. Du Pont de Nemours Company.





Contents

Hamilton Company Instrument Warranty	-i
Chapter 1 –	
Introduction	1-1
Chapter 2 –	
Specifications	2-1
Chapter 3 –	
Hardware	3-1
Mounting hole locations	3-1
Valve selection and installation	3-2
Valve Installation	3-2
Syringe selection and installation	3-4
Syringe Installation	3-5
Syringe drive thumbscrew location	3-6
Chapter 4 –	
Communications	4-1
DIP switch configuration table	4-3
Jumper configuration	4-4
Address switch	4-4
RS-232/485 connections	4-5
RS-485 connection	4-7
RS-485 communication termination with external resistors	4-8
CAN communications connections	4-9
Chapter 5 –	
Programming Introduction	5-1
Definitions	5-2
Steps	5-2
Standard Resolution	5-2
High Resolution	5-2
Motor Steps	5-2
Backoff Steps	5-2
Return Steps	5-2
Start Velocity	5-2
Maximum Velocity	5-2
Stop Velocity	5-3
Controlling Device	5-3
Commands	5-3
Command Buffer	5-3
Action Commands	5-3
Control Commands	5-3
Async Commands	5-3
Query Commands	5-3
Command String	5-4
Response string	5-4

**Chapter 6 –**

General Description	6-1
Error status	6-2
Pump status	6-2
Command buffer execution	6-3
Syringe motion	6-3
PSD/4 Initialization	6-4

Chapter 7 –

Protocol Information	7-1
RS232 Communication	7-2
Parameter settings	7-2
RS232/RS485 protocols	7-2
PSD/4 RS232 address	7-3
Terminal protocol	7-4
Terminal protocol data block	7-4
Standard protocol	7-5
Standard protocol data block	7-5
Checksum	7-5
Sequence data	7-6
Error detection and recovery	7-6
Can protocol	7-8
CAN message	7-8
CAN message identifier	7-8
RTR	7-8
CAN data length	7-8
CAN data bytes	7-8
PSD/4 CAN address	7-9
PSD/4 CAN data block	7-9
Message types	7-10
Message class assignment	7-10
PSD/4 CAN acknowledgement	7-11
PSD/4 CAN communications overview	7-11
PSD/4 CAN control commands	7-12
PSD/4 CAN query commands	7-13
PSD/4 CAN Examples	7-14
Async Command Message Type example	7-14
Action Command Message Type example	7-17
PSD/4 CAN Control Command Message Type example:	
Terminate	7-19
PSD/4 CAN Control Command Message Type example:	
Reset	7-22
Multi-Message Start Message Type example	7-23
Multi-Message Continuation Message Type example	7-26
PSD/4 CAN Query Command example	7-30

**Chapter 8 –**

ASCII Command Set - Detailed	8-1
Control commands	8-2
Initialization commands	8-3
Syringe commands	8-5
Valve commands	8-9
Action commands	8-10
Motor control commands	8-14
Async commands	8-19
Query commands	8-20
Command summary	8-22

Chapter 9 –

h Factor Command	9-1
h factor command details	9-3
Enable/disable h factor commands	9-3
Syringe commands	9-3
Valve commands	9-4
PSD/4 commands	9-5
Query commands	9-6
Syringe query commands	9-6
Valve query commands	9-7
System query commands	9-8
Command summary	9-9
Valve input/output port locations	9-13

Appendix A –

Contacting Hamilton Company	1
In the United States and Canada	A-1
Outside the U.S. and Canada	A-1
In Switzerland	A-1





Chapter 1 Introduction

The PSD/4 (Precision Syringe Drive/4) is a compact syringe pump designed to perform all liquid handling operations including dispensing, serial dispensing and diluting.

The precision manufactured body provides a rigid platform for system components resulting in a reduction of overall system hysteresis. This combined with a self-lubricating, wear compensating lead nut provide unsurpassed syringe plunger positioning. Each PSD/4 is provided with a National Institute of Standards and Technology (N.I.S.T.) traceable performance test report performed on that module, assuring outstanding accuracy and precision.

The module can be fitted with a choice of syringes and valves to meet even the most demanding applications.

Operating on 24VDC, the PSD/4 can be used either as a single syringe pump or connected in series to form a bank of up to sixteen modules. Control is available in either an RS232 or CAN (Control Area Network) format.

The fluid contact surfaces of the PSD/4 are chemically inert materials, such as, PTFE, FEP, CTFE, ceramic, and glass.





Chapter 2 Specifications

The following tables define the operational specifications for the PSD/4 syringe pump module.

Specifications	
Resolution	Selectable 3,000 standard resolution 24,000 high resolution
Speed (full stroke)	1 second to 20 minutes
Syringe drive mechanism	DC Stepper Motor, self lubricating and wear compensating lead nut/screw, home sensor, 30 mm stroke length
Valve drive	Direct Drive DC Stepper motor with 15° resolution and optical encoder
Valve drive speed	250 ms per 120° rotation
Power requirements	24 VDC $\pm$ 10%
Power rating	1.5 A (peak)
Weight	2.6 lbs. (1.2 kg)
Dimensions (refer to figure 3-1)	Height: 5.00" Width: 1.75" Depth: 4.20"
Storage temperature	-20° to 65° C
Operating range	Temperature: 15° to 40° C Humidity: 20% to 95% R.H. non-condensing
Communication	CAN, RS232, RS485, via ASCII commands
Baud rate	9600 or 38400 (RS232, RS485) 100K or 125K (CAN)
Additional Regulatory Compliance Information	
Pollution degree	2
Installation category	II
Altitude	2000 m

Table 2-1. Specifications

Indoor Operation and Use Only



Syringe size (µL)	Percent stroke	Accuracy (±)	Precision
12.5	5% ≤ Stroke < 30%	3.0%	2.0%
	30% ≤ Stroke	2.0%	0.2%
25	5% ≤ Stroke < 30%	3.0%	2.0%
	Stroke = 30%	1.5%	0.2%
	Stroke = 100%	1.0%	0.2%
50	5% ≤ Stroke < 30%	2.0%	1.0%
	Stroke = 30%	1.0%	0.2%
	Stroke = 100%	1.0%	0.1%
100/125	5% ≤ Stroke < 30%	2.0%	1.0%
	Stroke = 30%	1.0%	0.2%
	Stroke = 100%	1.0%	0.1%
250	5% ≤ Stroke < 30%	1.5%	0.5%
	Stroke = 30%	1.0%	0.2%
	Stroke = 100%	1.0%	0.05%
500	1% ≤ Stroke < 5%	3.0%	1.5%
	5% ≤ Stroke < 30%	1.5%	0.5%
	Stroke = 30%	1.0%	0.2%
	Stroke = 100%	1.0%	0.05%
1 ml & larger	1% ≤ Stroke < 5%	3.0%	1.5%
	5% ≤ Stroke < 30%	1.2%	0.5%
	Stroke = 30%	1.0%	0.1%
	Stroke = 100%	1.0%	0.05%
1) Reference Temperature: 21.5 ± 2.5°C 2) Default Back-off & Return Steps are used.			

Table 2-2. PSD/4 accuracy and precision (with syringe)



Chapter 3 Hardware

Mounting hole locations

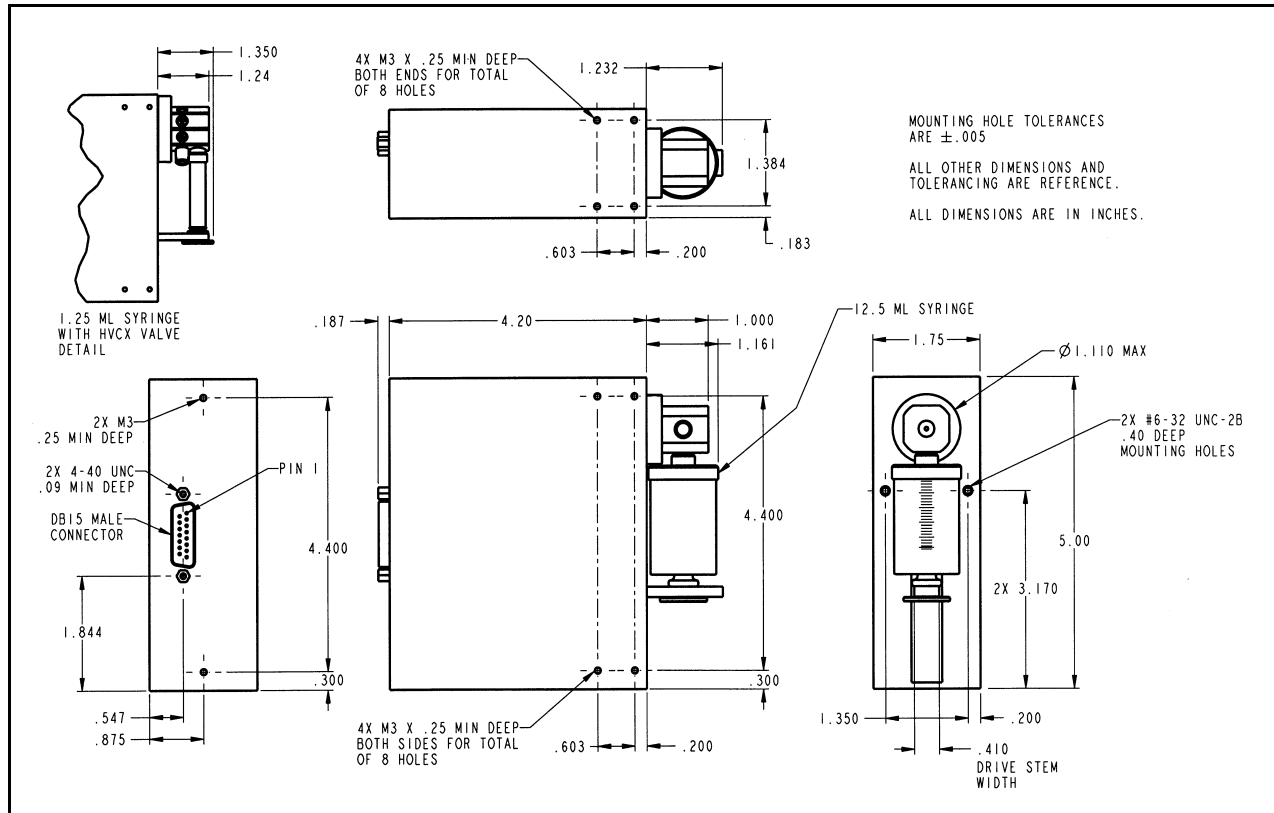


Figure 3-1: Drawing of mounting hole locations



Valve selection and installation

Standard Valves		
Valve Configuration	HV Part number	Fluid Contact Materials
Three port Y-valve	7427-01	CTFE, PTFE
T- port	7990-01	CTFE, PTFE
Ceramic Valves		
Valve Configuration	HV Part number	Fluid Contact Materials
Three port Y-valve	8778-01	PTFE, Ceramic
T-port	8063-01	PTFE, Ceramic
Four port wash valve	9234-01	PTFE, Ceramic
Distribution valves		
Valve Configuration	HV Part number	Fluid Contact Materials
Three port distribution	7991-01	PTFE, Ceramic
Four port distribution	7992-01	PTFE, Ceramic
Six port distribution	9998-01	PTFE, Ceramic
Eight port distribution	7993-01	PTFE, Ceramic

Table 3-1. Valve selection and installation

Valve Installation

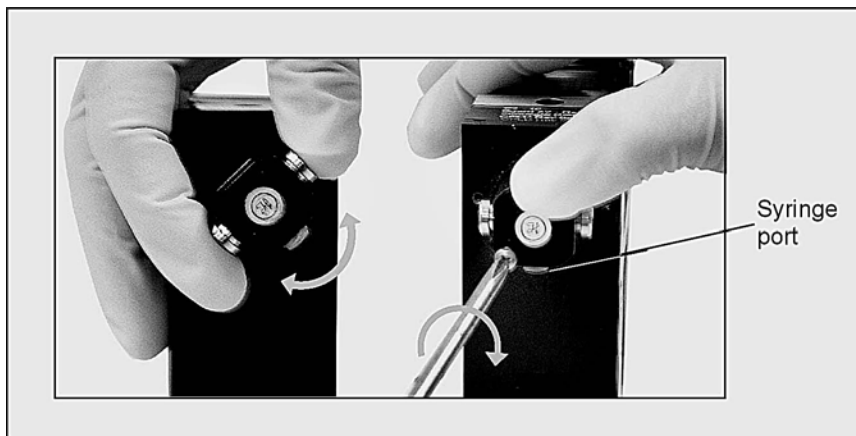


Figure 3-2: Valve installation



To install the valve:

1. Insert valve into PSD/4.
2. Rotate valve until the drive stems engage.
3. Continue to rotate valve until the syringe port is aligned with the plunger drive stem, facing down.
4. When the alignment pins engage, press the valve firmly against the PSD/4.
5. Tighten mounting screws no greater than 40 inch ounces.



Syringe selection and installation

Volume	Teflon-tipped Syringes	UHMWPE-tipped Syringes
12.5 µL	5495-10	
25 µL	5495-15	8250-15
50 µL	5495-20	8250-20
100 µL	5495-22	8250-22
125 µL	5495-25	8250-25
250 µL	5495-30	8250-30
500 µL	5495-35	8250-35
1.0 ml	5495-37	8250-37
1.25 ml	5495-40	8250-40
2.5 ml	5495-45	8250-45
5 ml	5495-50	8250-50
12.5 ml	5495-55	

Table 3-2. Syringe Part Numbers for use with the PSD/4

PSD4 Syringe Compatibility					
Syringe	Volume	HV	HVC HVCXI (6)	HVXI (8)	HVCXI (4 & 8)
1705	50 µL	Y	Y	Y	Y
1710	100 µL	Y	Y	Y	Y
1725	250 µL	Y	Y	Y	Y
1750	500 µL	Y	Y	Y	Y
1001	1.0 ml	Y	Y	Y	Y
1002	2.5 ml	Y	Y	N	Y
1005	5 ml	Y	Y	N	N
1010	10 ml	Y	Y	N	N
1025	25 ml	Y	Y	N	N
Note: 1. Y = Yes, the syringe works OK with the valve indicated 2. N = No, the syringe does not work with the valve indicated					

Table 3-3. PSD4 Syringe Compatibility



Syringe Installation



To install the syringe:

1. Position syringe plunger to the center of the stroke (see figure 3-3).

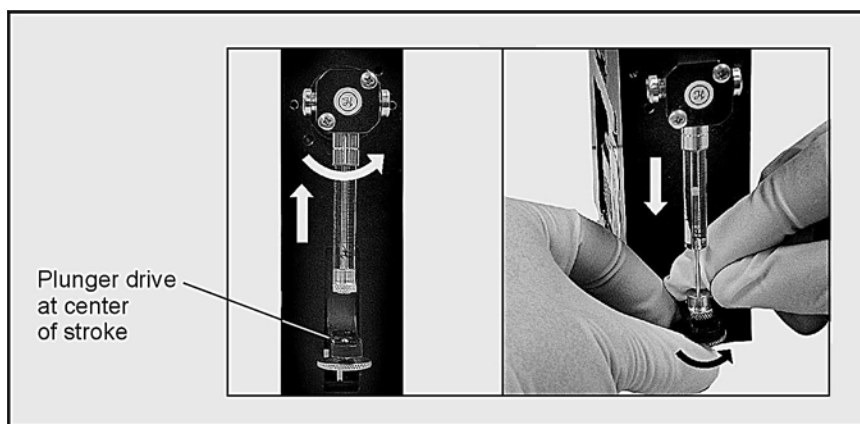


Figure 3-3: Syringe installation



Position the plunger drive stem in the middle to bottom of stroke range for syringe installation or removal.

2. Insert the luer end of the syringe into the valve and rotate until tight.
3. Pull the plunger down until it reaches the syringe.
4. Position the plunger so that the plunger button sets into the channel of the drive stem.
5. Hold the plunger and tighten the thumbscrew into the plunger.
6. Initialize the PSD/4.
7. Re-tighten syringe into valve.



Syringe drive thumbscrew location

There may be a need to relocate the thumbscrew on the syringe depending on the syringe/valve combination being used.



To Remove:

1. Pry the E-clip away from the thumbscrew shaft.
2. Remove E-clip and thumbscrew.



To install:

1. Drop E-clip into drive stem position.
2. Insert thumbscrew into position.
3. Apply a lateral pressure on the thumbscrew against the open face of the E-clip until the E-clip engages.



The syringe is required to be installed parallel to the face of the PSD/4 or damage will result to the syringe and/or valve.

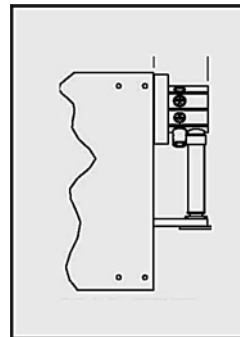


Figure 3-4: Syringe drive thumbscrew location



Chapter 4 Communications

DIP switch configuration table
Jumper configuration
Address switch
RS-232/485 connections
RS-485 connection
RS-485 communication termination with external resistors
CAN communications connections

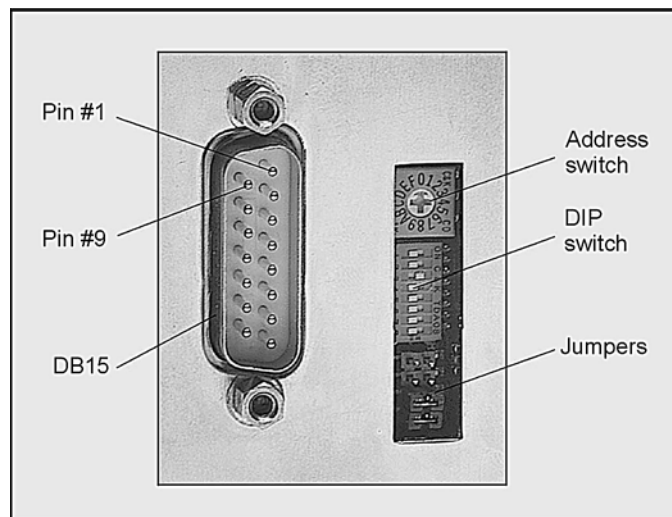


Figure 4-1: PSD/4 rear view



Pin	Function	Remark
1	24 VDC	
2	RS-232 TxD line	Output data
3	RS-232 RxD line	Input data
4	RS-232 HI line	Line is high with power on
5	CAN high signal line	
6	CAN low signal line	
7	Auxiliary Input #1	Digital level
8	Auxiliary Input #2	Digital level
9	Ground	Power and Logic
10	Ground	Power and Logic
11	RS-485 A line	
12	RS-485 B line	
13	Auxiliary Output #1	Digital level
14	Auxiliary Output #2	Digital level
15	Auxiliary Output #3	Digital level

Table 4-1. DB-15 Connector Pin Assignments



DIP switch configuration table

Switch Position Description	Details	Default	Switch Circuit							
			1	2	3	4	5	6	7	8
Syringe Overload Detection	Enabled	X	OFF	-	-	-	-	-	-	-
	Disabled		ON	-	-	-	-	-	-	-
EEPROM AutoStart/Self Test	Disabled	X	-	OFF	-	-	-	-	-	-
	Enabled		-	ON (1)	-	-	-	-	-	-
Baud Rate	9600 baud 100 Kbaud for CAN	X	-	-	OFF	-	-	-	-	-
	38400 baud 125 Kbaud for CAN		-	-	ON	-	-	-	-	-
Force Boot	Enabled		-	-	-	ON	ON	ON	-	-
3 - Port Y Valve		X	-	-	-	OFF	OFF	OFF	-	-
T - Port Valve			-	-	-	ON	OFF	OFF	-	-
3 - Port Distribution Valve			-	-	-	OFF	ON	OFF	-	-
8 - Port Distribution Valve			-	-	-	ON	ON	OFF	-	-
4 - Port Distribution Valve 4 - Port Wash Valve			-	-	-	OFF	OFF	ON	-	-
RS-485 Communication Termination	Single unit, first or last unit in chain	X	-	-	-	-	-	-	ON (2)	ON (3)
	Non-end unit in chain		-	-	-	-	-	-	OFF	OFF
Notes: 1. Self-Test actuated with address switch set to "F", Address Switch set to "0-E" executes Command Strings stored in EEPROM locations 0-14 2. RS-485-A 3. RS-485-B 4. A dash, "-", represents a switch circuit that has no effect on the associated configuration.										

Table 4-2. Dip switch configuration



Jumper configuration

Designations	Settings
In System Programming	1-2, 3-4
Normal Operation	5-6, 7-8

Table 4-3. Jumper configuration

Please note that the jumpers are factory set. Any alteration to the original settings (Normal Operation) will result in miscommunication, errors, and/or damage to the PSD/4.

Address switch

A 16 position rotary switch is provided for setting the address position of each module for RS232 communication.

Address Switch	PSD/4 Number	Address Character (Terminal & Standard Protocol)
0	1	1
1	2	2
2	3	3
3	4	4
4	5	5
5	6	6
6	7	7
7	8	8
8	9	9
9	10	:
A	11	;
B	12	<
C	13	=
D	14	>
E	15	?
F	16	@

Table 4-4. Address Switch Settings



RS-232/485 connections

The following describes how the pump is connected for operation using an RS-232 controller, such as a PC serial port. See figure 4-2.

- The first pump is connected to an RS-232 port. See table 4-5.

PC Serial Port Connector		Pump #1 Connector	
Function	DB-9	Function	DB-15
TXD	2	RXD	2
RXD	3	TXD	3
CTS	4	HI	4
GND	5	GND	10

Table 4-5. RS-232 computer to pump #1 cable

- Successive PSD/4s are connected through pins 11 (RS-485 A) and 12 (RS-485 B) of the DB-15 connectors.
- Power is supplied to pins 1 (24 VDC) and 9 (GND) of the DB-15 connectors. No more than two devices should be connected in series to the same power line.
- The Address Switch is set such that the first PSD/4 is set to “0”, second to “1”, and so forth.
- Up to sixteen devices can be addressed from one controller port.
- The communication termination switches must be set on both the first and last units. The middle units are left open. External resistors can be used instead of the communication switches on the last device in an RS-485 chain. If the resistors are used, the termination switches are not required.

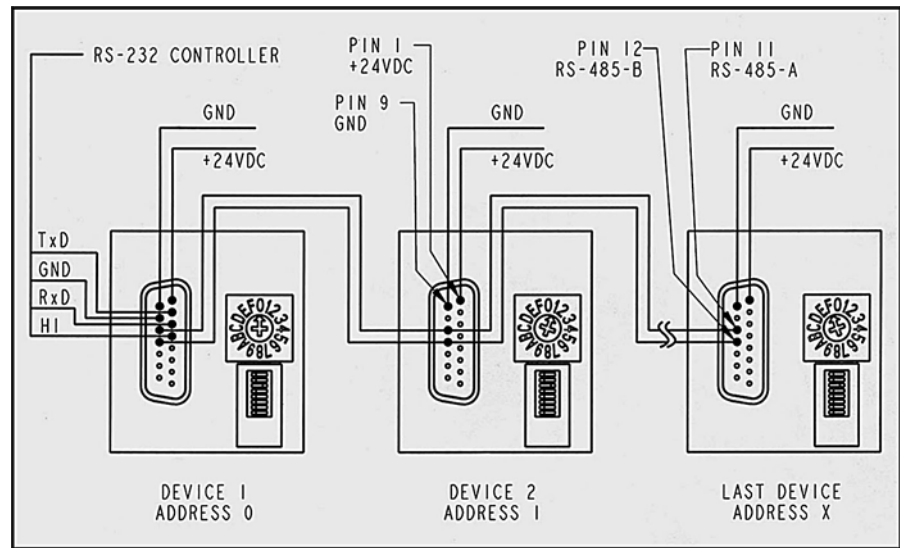


Figure 4-2: RS-232/485 connections



RS-485 connection

The following describes how the pump should be hooked-up to operate from an RS-485 controller. See figure 4-3.

- PSD/4s are connected through pins 11 (RS-485 A) and 12 (RS-485 B) of the DB-15 connectors to the controller and/or other devices in the chain.
- Power is supplied to pins 1 (24 VDC) and 9 (GND) of the DB-15 connectors. No more than two devices should be connected in series to the same power line.
- The Address Switches is set such that the first PSD/4 is set to “0”, second to “1”, and so forth.
- Up to sixteen devices can be addressed from one controller port.
- The communication termination switches are required to be set on the last unit. The first and middle units are left open.
- External resistors can be used instead of the communication switches on the last device in an RS-485 chain. If the resistors are used, the termination switches are not required.

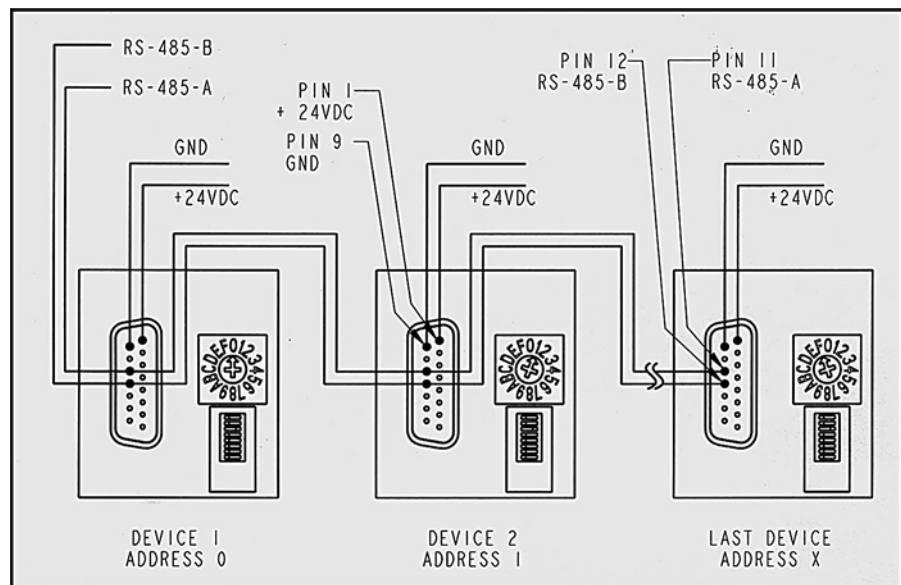


Figure 4-3: RS-485 connections



RS-485 communication termination with external resistors

External resistors can be used (see figure 4-4), instead of the communication switches on the last device in an RS-485 chain. If the resistors are used, the termination switches are not required.

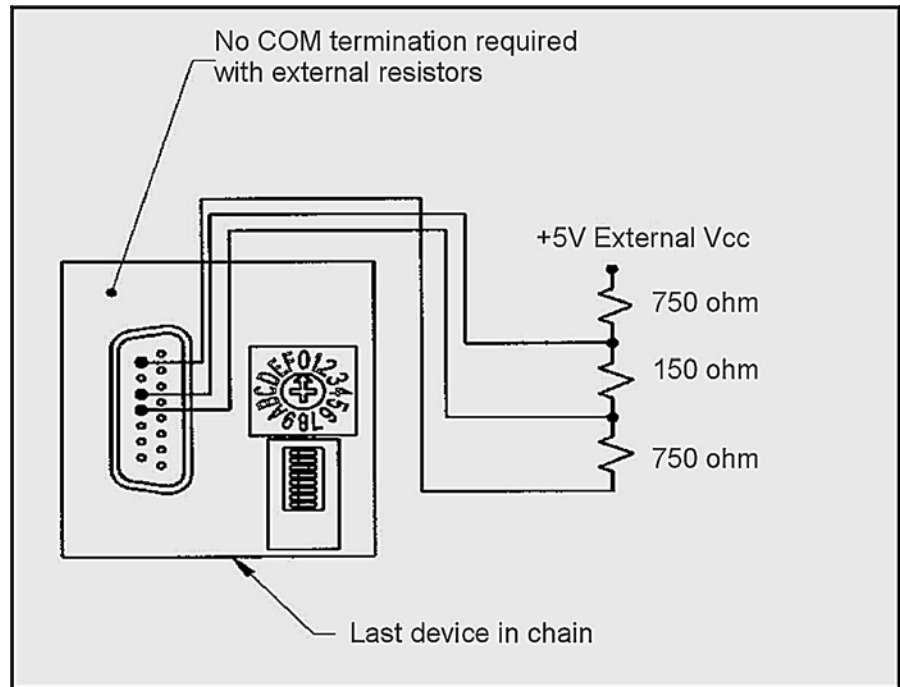


Figure 4-4: RS-485 termination with external resistors



CAN communications connections

The following describes how the pump is connected when operating from a CAN (Controller Area Network) controller (see figure 4-5).

- PSD/4s are connected through pins 5 (CAN high) and 6 (CAN low) of the DB-15 connectors to the CAN controller and/or other devices in the chain.
- Power is supplied to pins 1 (24 VDC) and 9 (GND) of the DB-15 connectors. No more than two devices should be connected in series to the same power line.
- The Address Switch is set such that the first PSD/4 is set to “0”, second to “1”, and so forth.
- Up to sixteen devices can be addressed from one controller port.
- The communication termination switches are not needed for CAN hook-up.
- CAN termination is not provided by the PSD/4.

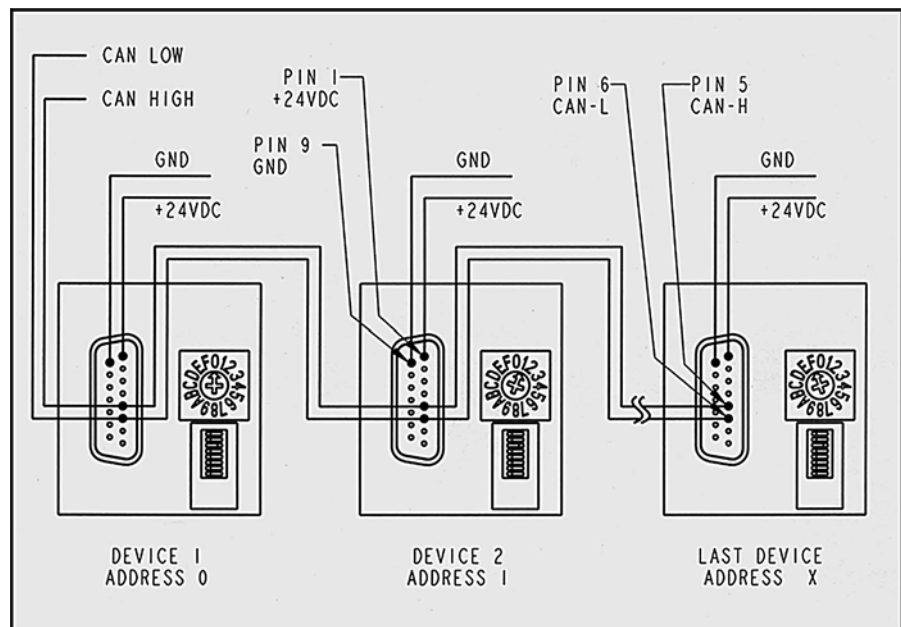


Figure 4-5: CAN connections





Chapter 5

Programming Introduction

Definitions

Steps

Standard Resolution

High Resolution

Motor Steps

Backoff Steps

Return Steps

Start Velocity

Maximum Velocity

Stop Velocity

Controlling Device

Commands

Command Buffer

Action Commands

Control Commands

Async Commands

Query Commands

Command String

Response string



Definitions

Steps

Steps are the number of stopping positions available for use with the PSD/4.

Standard Resolution

Standard Resolution is the default resolution for the PSD/4. In Standard Resolution, the PSD/4 has 3000 Steps per full stroke.

High Resolution

High Resolution is an additional mode the PSD/4 supports that allows 24000 Steps per full stroke.

Motor Steps

Motor steps are the physical number of motor steps used by the syringe drive. The PSD/4 has 6000 motor steps per full stroke.

Backoff Steps

Backoff Steps refers to the number of Steps the syringe motor moves down after the initial motor stall during an initialization command. The syringe backoff prevents syringe tip compression from adversely effecting accuracy and precision.

Return Steps

Return Steps refers to the number of Steps the syringe motor uses to compensate for mechanical backlash, which increases syringe accuracy and precision. Each downward movement of the syringe drive travels an extra Return Steps steps, and is immediately followed by an upward movement of Return Steps steps.

Start Velocity

The velocity in Motor Steps per second at which a syringe move starts. The Start Velocity used for a given move is never greater than the Maximum Velocity for that move.

Maximum Velocity

The maximum velocity in Motor Steps per second the syringe motor may attempt to reach.



Stop Velocity

The velocity in Motor Steps per second at which a syringe move ends. The Stop Velocity used for a move in the down direction is equal to the Start Velocity for that move. The Stop Velocity used for a given move is never less than the Start Velocity and never greater than the Maximum Velocity for that move.

Controlling Device

The system used to communicate with the PSD/4.

Commands

Commands are the primary communications syntax used by the PSD/4. All commands are a single character followed by a numeric parameter. If the parameter is omitted, it is assumed to be zero. Some commands do not require a parameter, and therefore ignore the parameter.

Command Buffer

The Command Buffer is a list of zero or more action commands to be executed by the PSD/4.

Action Commands

Action Commands consist of the set of commands that may be stored in the Command Buffer.

Control Commands

Control Commands consist of those commands that are used to start or resume execution of the Command Buffer.

Async Commands

Async Commands consist of those commands that affect the PSD/4 while the Command Buffer is being executed.

Query Commands

Query Commands consist of those commands that are used to return information about the PSD/4 to the Controlling Device.



Command String

A valid Command String consists of one of the following:

- a. Nothing
- b. An Async Command
- c. A Query Command
- d. A Control Command
- e. One or more Action Commands
- f. One or more Action Commands followed by a Control Command

In cases a, e, and f above:

- 1. The Command Buffer is cleared before the Command String is processed.
- 2. The Action Commands (if any) in the Command String are placed into Command Buffer.
- 3. A Control Command that starts execution of the Command Buffer starts execution from the beginning of the Command Buffer.

In cases b, c, and d above:

- 1. The command is processed immediately.

Response string

A Response String consists of data being returned from the PSD/4 to the Controlling Device. The first byte of all Response Strings is the Pump Status. The term Response String does not refer to any of the protocol information that accompanies the response data.



Chapter 6

General Description

Error status
Pump status
Command buffer execution
Syringe motion
PSD/4 Initialization



Error status

The Error Status is the last error reported by the PSD/4.

Error	Description
0	No error.
1	Initialization error. The valve or syringe failed to initialize.
2	Invalid command. The command received was not recognized as a valid command.
3	Parameter out of range. The parameter specified for a command was not within the range of values acceptable for that command.
4	Too many loops.
6	EEPROM error. The EEPROM is unable to store information properly.
7	Syringe not initialized. A syringe move was attempted when the syringe was not initialized.
9	Syringe overload. The syringe motor stalled during a move.
10	Valve overload. The valve motor stalled during a move.
11	Syringe move not allowed. A syringe move was attempted when the valve was in the bypass position.
15	PSD/4 busy error. The PSD/4 is unable to process the command because the PSD/4 is busy executing the command buffer.

Pump status

The status of PSD/4 is available as the Pump Status. The PSD/4 Pump Status is a single byte that indicates the ready / busy status of the PSD/4 and the Error Status. The PSD/4 indicates a status of busy while it is executing commands from the Command Buffer, and ready otherwise. The Error Status contains the last error reported by the PSD/4. The PSD/4 clears the Error Status after reporting it.

The 8-bit Pump Status is defined as follows:

Bit 7	Always 0
Bit 6	Always 1
Bit 5	1 if ready, 0 if busy.
Bit 4	Always 0
Bits 3 - 0	Error Status



The following Commands Strings, if received while the PSD/4 is busy, will result in a “pump busy” Error Status.

- a. Nothing
- b. A Control Command
- c. One or more Action Commands
- d. One or more Action Commands followed by a Control Command

Command buffer execution

The PSD/4 executes Action Commands in the Command Buffer in the order they were specified in the Command String. The PSD/4 continues to execute the commands in the Command Buffer until:

- a. No more commands remain to be executed in the Command Buffer.
- b. An error occurs executing a command.
- c. A Control Command is received to terminate execution.

Syringe motion

Based up on the programmed Start Velocity, Maximum Velocity, Stop Velocity, acceleration, and the number of Steps to move, the syringe motor will move in one of the following ways:

- a. The syringe motor starts, runs, and stops at the Maximum Velocity.
- b. The syringe motor accelerates from the Start Velocity to a velocity less than or equal to the Stop Velocity, then stops.
- c. The syringe motor accelerates from the Start Velocity to a velocity less than or equal to the Maximum Velocity, then decelerates to the Stop Velocity.
- d. The syringe motor accelerates from the Start Velocity to the Maximum Velocity, runs at the Maximum Velocity, then decelerates to the Stop Velocity.



PSD/4 Initialization

The PSD/4 performs the following sequence when an initialization command is executed with a valve configured.

- a. If the valve is not initialized, the valve will perform an initialization sequence. If the initialization fails, an Initialization Error will occur.
- b. The valve moves to the output position.
- c. The syringe moves upward until the motor stalls and the syringe home sensor is detected. If the motor does not stall or the home sensor is not detected, an Initialization Error will occur.
- d. The valve moves to the input position.
- e. The syringe moves down (Backoff Steps + Return Steps) steps.
- f. The valve moves to the output position.
- g. The syringe moves up Return Steps steps.

The PSD/4 performs the following sequence when an initialization command is executed with no valve configured.

- a. The syringe moves upward until the motor stalls and the syringe home sensor is detected. If the motor does not stall or the home sensor is not detected, an Initialization Error will occur.
- b. The syringe moves down (Backoff Steps + Return Steps) steps.
- c. The syringe moves up Return Steps steps.



Chapter 7

Protocol Information

- RS232 Communication
 - Parameter settings
- RS232/RS485 protocols
- PSD/4 RS232 address
- Terminal protocol
 - Terminal protocol data block
- Standard protocol
 - Standard protocol data block
 - Checksum
 - Sequence data
 - Error detection and recovery
- Can protocol
 - CAN message
 - CAN message identifier
 - RTR
 - CAN data length
 - CAN data bytes
 - PSD/4 CAN address
 - PSD/4 CAN data block
 - Message types
 - Message class assignment
 - PSD/4 CAN acknowledgement
 - PSD/4 CAN communications overview
 - PSD/4 CAN control commands
 - PSD/4 CAN query commands
 - PSD/4 CAN Examples
 - Async Command Message Type example
 - Action Command Message Type example
 - PSD/4 CAN Control Command Message Type example:
 - Terminate
 - PSD/4 CAN Control Command Message Type example:
 - Reset
 - Multi-Message Start Message Type example
 - Multi-Message Continuation Message Type example
 - PSD/4 CAN Query Command example



RS232 Communication

Parameter settings

Parameter	Setting
Baud rate	9600 or 38400
Data bits	8
Parity	None
Stop bit	1
Handshaking	None

Table 7-1. RS232 parameter settings

RS232/RS485 protocols

The PSD/4 supports two protocols across RS232/RS485, Standard Protocol and Terminal Protocol. The specific protocol to use is decided by the Controlling Device. The Terminal Protocol is the easiest to use from a terminal emulation program such as HyperTerm, but has no communications error checking. Standard Protocol is typically used by user-written applications that require better error checking and can automate the Checksum computation. The PSD/4 automatically detects which of these two protocols are being sent to it, and responds using the same protocol.

Data Blocks are the basic unit of communications between the Controlling Device and the PSD/4 when using Standard or Terminal Protocols. The Controlling Device initiates all communications to PSD/4s with a Data Block that contains a Command String, and the PSD/4 responds with a Data Block that contains a Response String (except when Broadcast addressed).



PSD/4 RS232 address

Each PSD/4 in a given RS232 / RS485 chain is required to have a unique address. The PSD/4 RS232 Address of each PSD/4 is selected by the address switch on the back of the PSD/4. The address switch is read by the PSD/4 at power-up only. The PSD/4 ignores all Data Blocks that contain a PSD/4 RS232 Address that does not correspond to its address switch setting (see table 7-2). PSD/4s will not send a response to any Data Block that contains a PSD/4 RS232 Address that addresses more than a single PSD/4. The address of the Controlling Device on an RS232/RS485 chain is hexadecimal 30, ASCII 0.

Address switch	1 PSD/4 Address		2 PSD/4s Address		4 PSD/4s Address		16 PSD/4s Address	
	Hex	ASCII	Hex	ASCII	Hex	ASCII	Hex	ASCII
0	31	1	41	A	51	Q	5F	-
1	32	2						
2	33	3						
3	34	4	43	C	55	U		
4	35	5						
5	36	6						
6	37	7	47	G				
7	38	8						
8	39	9			49	I		
9	3A	:						
A	3B	;						
B	3C	<						
C	3D	=	4D	M	5D	J		
D	3E	>						
E	3F	?						
F	40	@	4F	O				

Table 7-2. Address switch setting



Terminal protocol

The Terminal Protocol uses two standard characters (see table 7-3):

Notation	Name	ASCII code (decimal)	Description
<CR>	Carriage Return	13	The control character that terminates a Command String.
<LF>	Line Feed	10	The control character that terminates a response.

Table 7-3. Terminal protocol

Terminal protocol data block

The Terminal Protocol Data Block is the basic unit of communications between the Controlling Device and the PSD/4 when using the Terminal Protocol. A Terminal Protocol Data Block consists of the following:

`/<Address><Data><CR><LF>`

where:

/	Beginning of Data Block
<Address>	Controlling device or PSD/4 RS232 Address Character
<Data>	Command String or Response String
<CR>	Carriage Return
<LF>	Line Feed (optional for Controlling Device)



Standard protocol

The Standard Protocol uses two standard characters and one non-standard character (see table 7-4 and table 7-5):

Notation	Name	ASCII code (decimal)	Description
<STX>	Start of Text	02	The control character that indicates the beginning of a Data Block.
<ETX>	End of Text	03	The control character that indicates the end of a Data Block

Table 7-4. Standard protocol control characters

Notation	Name	ASCII code (decimal)	Description
<SYNC>	255	02	The character used to ensure a valid character has been transmitted. This character does NOT indicate the beginning a Data Block

Table 7-5. Non-standard protocol control characters

Standard protocol data block

The Standard Protocol Data Block is the basic unit of communications between the Controlling Device and the PSD/4 when using the Standard Protocol. A Standard Protocol Data Block consists of the following:

<SYNC><STX><Address><Sequence><Data><ETX><Checksum>

where:

<SYNC>	Valid serial character
<STX>	Beginning of Data Block
<Address>	Controlling Device or PSD/4 RS232 Address
<Sequence>	Sequence Data (Command Strings ONLY)
<Data>	Command String or Response String
<ETX>	End of Data Block
<Checksum>	Checksum

Checksum

The Checksum for a Data Block consists of the bitwise exclusive OR (XOR) of the bytes in the Data Block from the STX to the ETX, inclusive. A Data Block received with a Checksum that matches the computed Checksum is considered to be received successfully. A Data Block received with an invalid Checksum is ignored.



Sequence data

The Sequence Data is included in communications from the Controlling Device to the PSD/4 only. The Sequence Data consists of two parts, the Sequence Number and the Repeat Bit. The rest of the Sequence Data is fixed. The Sequence Field is defined as follows:

Bit 7:	Set to 0
Bit 6:	Set to 0
Bit 5:	Set to 1
Bit 4:	Set to 1
Bit 3:	Repeat Bit
Bits 2 - 0:	Sequence Number

Error detection and recovery

The Checksum, Sequence Number, and Repeat Bit facilitate the detection and recovery of communication errors between the Controlling Device and the PSD/4. This is performed in the following fashion:

When a Data Block with a valid Checksum is received by the PSD/4, the PSD/4 examines the Repeat Bit. If the Repeat Bit is zero, the PSD/4 will process the Command String, transmit a Data Block containing a Response String, and save the Sequence Number as the Sequence Number of the last successfully received Data Block. If the Repeat Bit is one, the PSD/4 will compare the received Sequence Number with the Sequence Number of the last successfully received Data Block. If the Sequence Numbers match, the PSD/4 will ignore the Command String and retransmit its last response. If the Sequence Numbers do not match, the PSD/4 processes the Command String as if the Repeat Bit was set to zero.

The following two examples illustrate how errors are handled and recovered from using the Sequence Number and Repeat Bit.

Case #1:

- a. The Controlling Device transmits a Data Block with the Sequence Number set to 3, Repeat Bit set to 0.
- b. The PSD/4 receives the Data Block with a valid Checksum. Since the Repeat Bit is set to 0, the PSD/4 processes the Command String and transmits a Data Block in response to the Command String. The PSD/4 saves the Sequence Number 3 as the last successfully received Data Block.
- c. The Controlling Device receives the Data Block from the PSD/4 with a valid Checksum.



- d. The Controlling Device transmits a Data Block with the sequence number set to 4, Repeat Bit set to 0.
- e. The PSD/4 receives the Data Block with an invalid Checksum. The PSD/4 ignores the Data Block.
- f. The Controlling Device does not receive a Data Block from the PSD/4 within a reasonable amount of time (100 milliseconds).
- g. The Controlling Device transmits the Data Block again with the Sequence Number set to 4, Repeat Bit set to 1.
- h. The PSD/4 receives the Data Block with a valid Checksum. Since the Repeat Bit is set to 1, the PSD/4 compares the received Sequence Number (4) with the Sequence Number of the last successfully received Data Block (3). Since they do not match in this case, the PSD/4 processes the Command String and transmits a Data Block in response to the Command String. The PSD/4 saves the Sequence Number 4 as the Sequence Number of the last successfully received Data Block.

Case #2

- a. The Controlling Device transmits a Data Block with the Sequence Number set to 2, Repeat Bit set to 0.
- b. The PSD/4 receives the Data Block with a valid Checksum. Since the Repeat Bit is set to 0, the PSD/4 processes the Command String and transmits a Data Block in response to the Command String. The PSD/4 saves the Sequence Number 2 as the Sequence Number of the last successfully received Data Block.
- c. The Controlling Device receives the Data Block with an invalid Checksum.
- d. The Controlling Device transmits the last Data Block with the Sequence Number set to 2, Repeat Bit set to 1.
- e. The PSD/4 receives the Data Block with a valid Checksum. Since the Repeat Bit is set to 1, the PSD/4 compares the received Sequence Number (2) with the Sequence Number of the last successfully received Data Block (2). Since they match in this case, the PSD/4 ignores the Command String, and re-transmits the last Data Block.



Can protocol

CAN message

A CAN Message consists of the following:

11 Bits	CAN Message Identifier
1 Bit	RTR
4 Bits	CAN Data Length
8 Bytes	CAN Data Bytes

CAN message identifier

The 11-bit CAN Message Identifier is further defined by the PSD/4 as follows:

Bit 10	Message Direction
Bits 9 - 7	Message Class
Bits 6 - 3	PSD/4 CAN Address
Bits 2 - 0	Message Type

Message Direction

This 1-bit field indicates the direction of communications. A value of 1 indicates the message is from the PSD/4 to the Controlling Device. A value of 0 indicates the message is from the Controlling Device to the PSD/4.

Message Class

This 3-bit field indicates the Message Class. The Message Class is set to 1 for Message Class Assignment requests and responses. The Message Class for subsequent communications is determined by the Message Class Assignment response.

Message Type

This 3-bit field indicates the type of message being transmitted.

RTR

This 1-bit field must be set to 0.

CAN data length

This 4-bit field indicates the number of CAN Data Bytes. This value must be between 0 and 8, inclusive.

CAN data bytes

This portion of the CAN Message contains the data being transferred between the PSD/4 and the Controlling Device.



PSD/4 CAN address

Each PSD/4 on a CAN network is required to have a unique address. The PSD/4 CAN Address of each PSD/4 is selected by the address switch on the back of the PSD/4. The address switch is read by the PSD/4 at power-up only. The PSD/4 ignores all Data Blocks that contain a PSD/4 CAN Address that does not match its address switch setting (see table 7-6). The address of the Controlling Device on a CAN network is binary 0000.

PSD/4 Address Switch Setting	PSD/4 CAN Address (binary)
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Table 7-6. PSD/4 CAN address

PSD/4 CAN data block

The CAN bus is limited to 8 CAN Data Bytes per CAN Message. The PSD/4 can receive data from, and transmit data to, the Controlling Device that exceeds the 8 data bytes in a single CAN Message. The process by which this is performed is called “Multi-Message”. A PSD/4 CAN Data Block is defined as all data contained in either a single CAN Message (total length less than or equal to 8 bytes), or all bytes contained in a Multi-Message (total length greater than 8 bytes).



Message types

Message Type 0 indicates that the CAN Message is the end of a PSD/4 CAN Data block that contains an Async Command. This message type is also used by the Message Class Assignment response.

Message Type 1 indicates that the CAN Message is the end of a PSD/4 CAN Data Block that contains Action Commands.

Message Type 2 indicates that the CAN Message is the end of a PSD/4 CAN Data Block that contains a PSD/4 CAN Control Command. This message type is also used by the Message Class Assignment request.

Message Type 3 indicates that the CAN Message is the beginning of a Multi-Message.

Message Type 4 indicates that the CAN Message is a continuation of a Multi-Message.

Message Type 6 indicates that the CAN Message is the end of a PSD/4 CAN Data Block that contains a PSD/4 CAN Query Command.

Message class assignment

In order to use the CAN protocol with the PSD/4, Message Class Assignment must be performed. Upon power-up or reset, the PSD/4 will transmit Message Class Assignment requests at 100 millisecond intervals until it receives a Message Class Assignment response.

Message Class Assignment request (see table 7-7):

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
001	Message Class	Message Class Assignment
0000	Controlling Device Address	
010	Message Type	Message Class Assignment Request
RTR and CAN Message Length		
0	RTR	
0000	Length	No data in request

Table 7-7. Message Class Assignment request



Message Class Assignment response (see table 7-8):

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
001	Message Class	Message Class Assignment
0000	PSD/4 CAN Address	(This value is for example only)
000	Message Type	Message Class Assignment response
RTR and CAN Message Length		
0	RTR	
0010	Length	Two bytes of data in response
CAN Data		
0010	Byte #1	Message Class Assigned to PSD/4 (for example)
0000	Byte #1	PSD/4 CAN Address (for example)
00100000	Byte #2	Same as CAN Data Byte #1 (for example)

Table 7-8. Message Class Assignment response

PSD/4 CAN acknowledgement

The PSD/4 confirms the receipt of PSD/4 CAN Data Blocks that contain message types 0, 1, and 2 with a PSD/4 CAN Acknowledgement. A PSD/4 CAN Acknowledgement for a given message type consists of a PSD/4 CAN Data Block of the same message type with a CAN Data Length of zero.

PSD/4 CAN communications overview

The response to at most one PSD/4 CAN Data Block of a given message type may be outstanding at a given time.

The response to a PSD/4 CAN Data Block with message type 0 is a PSD/4 CAN Acknowledgement only.

The response to a PSD/4 CAN Data Block with a message type of 1 or 2 is a PSD/4 CAN Acknowledgement. Upon completion of the request, the PSD/4 will transmit a Response String consisting of one byte of binary zero (note that all Response Strings are preceded by the Pump Status). Completion of the request can occur due to either the completion of the request, or to a terminate request.



The PSD/4 does not send a response to PSD/4 CAN Data Blocks with message types 3 or 4.

The response to PSD/4 CAN Data Blocks with message type 6 is the Response String for the Query.

PSD/4 CAN control commands

The CAN Protocol for the PSD/4 uses PSD/4 CAN Control Commands instead of the Control Commands used by the Standard and Terminal protocols. The PSD/4 CAN Control Commands are used to start, stop, or resume execution of the Command Buffer, as well as reset the PSD/4. The PSD/4 CAN Control Commands are represented by the ASCII digits 0 through 4, inclusive. PSD/4 CAN Control Commands are defined in table 7-9.

PSD/4 CAN Control Command	RS232 Protocol Command	Description
0	h30002	Reset PSD/4
1	R	Execute Command Buffer
2		Clear Command Buffer
3	X	Execute Command Buffer from Beginning
4	T	Terminate Execution

Table 7-9. PSD/4 CAN control commands



PSD/4 CAN query commands

The CAN Protocol for the PSD/4 uses PSD/4 CAN Query Commands instead of the Query Commands used by the Standard and Terminal protocols. The PSD/4 CAN Query Commands are used to return information about the PSD/4 to the Controlling Device. The PSD/4 CAN Query Commands are represented by the ASCII numbers 0 through 29, although not all are valid. The PSD/4 CAN Query Commands are defined in table 7-10.

PSD/4 CAN Query Command	RS232 Protocol Command	Description
0	?	Syringe position
1	?4	Syringe encoder position
4	?2	Syringe Maximum Velocity
5	?8	Actual Syringe Maximum Velocity
6	?1	Syringe Start velocity
7	?3	Syringe Stop velocity
8	?7	Actual Syringe Start Velocity
9	?9	Actual Syringe Stop Velocity
10	F	Command Buffer status
12	?12	Syringe Return Steps
13	?13	Digital input #1
14	?14	Digital input #2
23	&	Firmware version
24	?24	Syringe Backoff Steps
29	Q	Pump Status

Table 7-10. PSD/4 CAN query commands



PSD/4 CAN Examples

Async Command Message Type example

In this example, the Controlling Device prematurely terminates a syringe movement. Note that after the PSD/4 completes the terminate request, the pump status is sent using the Action Command Message Type.

1. The Controlling Device sends the following request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0110	Length	Six bytes of data in command
CAN Data		
01000001	Byte #1	ASCII A
00110011	Byte #2	ASCII 3
00110000	Byte #3	ASCII 0
00110000	Byte #4	ASCII 0
00110000	Byte #5	ASCII 0
01010010	Byte #6	ASCII R



2. The PSD/4 immediately sends its acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement

3. The Controlling Device sends the async terminate request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
000	Message Type	Async Command
RTR and CAN Message Length		
0	RTR	
0001	Length	One byte of data in command
CAN Data		
01010100	Byte #1	ASCII T



4. The PSD/4 immediately sends its acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
000	Message Type	Async Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement

5. When complete, the PSD/4 sends its response:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Response
CAN Data		
01100000	Byte #1	Pump Status (for example)
00000000	Byte #2	NULL Response String



Action Command Message Type example

In this example, the Controlling Device initializes the PSD/4 using the Action Command “ZR”.

1. The Controlling Device sends the following request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Two bytes of data in command
CAN Data		
01011010	Byte #1	ASCII Z
01010010	Byte #2	ASCII R

2. The PSD/4 immediately sends its acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement



3. When complete, the PSD/4 sends its response:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Response
CAN Data		
01100000	Byte #1	Pump Status (for example)
00000000	Byte #2	NULL Response String



PSD/4 CAN Control Command Message

Type example: Terminate

In this example, the Controlling Device prematurely terminates a syringe movement. Note that after the PSD/4 completes the terminate request, the pump status is sent using the PSD/4 CAN Control Command Message Type.

1. The Controlling Device sends the following request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0110	Length	Six bytes of data in command
CAN Data		
01000001	Byte #1	ASCII A
00110011	Byte #2	ASCII 3
00110000	Byte #3	ASCII 0
00110000	Byte #4	ASCII 0
00110000	Byte #5	ASCII 0
01010010	Byte #6	ASCII R



2. The PSD/4 immediately sends its acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement

3. The Controlling Device sends the PSD/4 CAN Control Command Terminate request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
010	Message Type	PSD/4 CAN Control Command
RTR and CAN Message Length		
0	RTR	
0001	Length	One byte of data in command
CAN Data		
00110100	Byte #1	ASCII 4



4. The PSD/4 immediately sends its acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
010	Message Type	PSD/4 CAN Control Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement

5. When complete, the PSD/4 sends its response:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
010	Message Type	PSD/4 CAN Control Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Response
CAN Data		
01100000	Byte #1	Pump Status (for example)
00000000	Byte #2	NULL Response String



**PSD/4 CAN Control
Command Message
Type example: Reset**

1. The Controlling Device sends the PSD/4 CAN Control Command Reset request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
010	Message Type	PSD/4 CAN Control Command
RTR and CAN Message Length		
0	RTR	
0001	Length	One byte of data in command
CAN Data		
00110000	Byte #1	ASCII 0

2. The PSD/4 immediately sends its acknowledgement prior to resetting:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
010	Message Type	PSD/4 CAN Control Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement



Multi-Message Start Message Type example

In this example, the Controlling Device cycles the PSD/4 one time using the Action Command “IP3000OD3000R”.

1. The Controlling Device sends the beginning of the request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
011	Message Type	Multi-Message Start
RTR and CAN Message Length		
0	RTR	
1000	Length	Eight bytes of data in command
CAN Data		
01001001	Byte #1	ASCII I
01010000	Byte #2	ASCII P
00110011	Byte #3	ASCII 3
00110000	Byte #4	ASCII 0
00110000	Byte #5	ASCII 0
00110000	Byte #6	ASCII 0
01001111	Byte #7	ASCII O
01000100	Byte #8	ASCII D



2. The Controlling Device sends the end of the request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0101	Length	Five bytes of data in command
CAN Data		
00110011	Byte #1	ASCII 3
00110000	Byte #2	ASCII 0
00110000	Byte #3	ASCII 0
00110000	Byte #4	ASCII 0
01010010	Byte #5	ASCII R

3. The PSD/4 immediately sends an acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement



4. When complete, the PSD/4 sends the response:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Response
CAN Data		
01100000	Byte #1	Pump Status (for example)
00000000	Byte #2	NULL Response String

**Multi-Message
Continuation
Message Type
example**

In this example, the Controlling Device cycles the PSD/4 one hundred times using the Action Command “IP3000OD3000G100R”.

1. The Controlling Device sends the beginning of the request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
011	Message Type	Multi-Message Start
RTR and CAN Message Length		
0	RTR	
1000	Length	Eight bytes of data in command
CAN Data		
01001001	Byte #1	ASCII I
01010000	Byte #2	ASCII P
00110011	Byte #3	ASCII 3
00110000	Byte #4	ASCII 0
00110000	Byte #5	ASCII 0
00110000	Byte #6	ASCII 0
01001111	Byte #7	ASCII O
01000100	Byte #8	ASCII D



2. The Controlling Device sends the middle portion of the request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
100	Message Type	Multi-Message Continuation
RTR and CAN Message Length		
0	RTR	
1000	Length	Eight bytes of data in command
CAN Data		
00110011	Byte #1	ASCII 3
00110000	Byte #2	ASCII 0
00110000	Byte #3	ASCII 0
00110000	Byte #4	ASCII 0
01000111	Byte #5	ASCII G
00110001	Byte #6	ASCII 1
00110000	Byte #7	ASCII 0
00110000	Byte #8	ASCII 0



3. The Controlling Device sends the end of the request:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0001	Length	One byte of data in command
CAN Data		
01010010	Byte #1	ASCII R

4. The PSD/4 immediately sends its acknowledgement:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0000	Length	Acknowledgement



5. When complete, the PSD/4 sends the response:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
001	Message Type	Action Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Response
CAN Data		
01100000	Byte #1	Pump Status (for example)
00000000	Byte #2	NULL Response String



PSD/4 CAN Query Command example

1. The Controlling Device sends the PSD/4 CAN Query Command Pump Status:

Value (binary)	Name	Description
CAN Message Identifier		
0	Message Direction	Controlling Device to PSD/4
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	PSD/4 CAN Address	(This value is for example only)
110	Message Type	PSD/4 CAN Control Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Two bytes of data in command
CAN Data		
00110010	Byte #1	ASCII 2
00111001	Byte #2	ASCII 9

2. The PSD/4 immediately sends the response:

Value (binary)	Name	Description
CAN Message Identifier		
1	Message Direction	PSD/4 to Controlling Device
010	Message Class	Message Class assigned to PSD/4 (This value is for example only)
0000	Controlling Device Address	
110	Message Type	PSD/4 CAN Query Command
RTR and CAN Message Length		
0	RTR	
0010	Length	Response
CAN Data		
01100000	Byte #1	Pump Status (for example)
00000000	Byte #2	NULL Response String



Chapter 8

ASCII Command Set - Detailed

- Control commands
- Initialization commands
- Syringe commands
- Valve commands
- Action commands
- Motor control commands
- Async commands
- Query commands
- Command summary



Control commands



See “PSD/4 CAN Examples” on page 7-14, for additional CAN Control Command information.

R - Execute Command Buffer

R executes the commands in the command buffer starting with the first unexecuted command in the command buffer.

When a Command String that consists of only an *R* is sent to the PSD/4 the PSD/4 will execute the command buffer starting with the first unexecuted command in the command buffer.

R is not required to execute Query Commands.

Command Example	Description
IA1500OA0G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.
R	Send R again, no action takes place.

Table 8-1. Execute command buffer - Example

X - Execute Command Buffer from Beginning

X executes the commands in the command buffer starting with the first command in the command buffer.

When a Command String that consists of only an *X* is sent to the PSD/4 the PSD/4 will execute the command buffer from the beginning.

Command Example	Description
IA1500OA0G8X	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.
X	Send <i>x</i> again, The Command String repeats from the beginning.

Table 8-2. Execute command buffer from beginning - Example



Initialization commands



See “PSD/4 CAN Examples” on page 7-14, for additional CAN Initialization Command information.

Zx - Initialize PSD/4, Assign Valve Output to Right

Z initializes the syringe to the home position and sets valve output position to the right side of the PSD/4 (as viewed from the front of the PSD/4).

Parameter - $x = 0$ initializes at full plunger force; $x = 1$ initializes at half plunger force, or speed where $10 \leq x \leq 40$.

All of the valves used on the PSD/4 have a designated input and output port for PSD/4 initialization. Please refer to table 9-3 for input/output port designations.

Command Example	Description
ZR	Initialize the syringe to the home position and set valve output position to the right side of the PSD/4.

Table 8-3. Initialize PSD/4, assign valve output to right - Example

Yx - Initialize PSD/4, Assign Valve Output to Left

Y initializes the syringe drive to the home position and sets valve output to the left side of the PSD/4 (as viewed from the front of the PSD/4).

Parameter - $x = 0$ initializes at full plunger force; $x = 1$ initializes at half plunger force, or speed where $10 \leq x \leq 40$.

All of the valves used on the PSD/4 have a designated input and output port for PSD/4 initialization. Please refer to table 9-3 for input/output port designations.

Command Example	Description
YR	Initialize the syringe drive to the home position and set valve output to the left side of the PSD/4.

Table 8-4. Initialize PSD/4, assign valve output to left - Example

Wx - Initialize PSD/4, Configure for No Valve

W initializes the syringe for a PSD/4 without a valve drive.

Parameter - $x = 0$ initializes at full plunger force; $x = 1$ initializes at half plunger force, or speed where $10 \leq x \leq 40$.



Once the *W* command is issued to a PSD/4, valve commands will be ignored until the power is cycled to the PSD/4 or the valve drive is re-enabled.

Command Example	Description
WR	Initialize and configure unit for no valve.

Table 8-5. Initialize PSD/4, configure for no valve - Example



Syringe commands

z - Set Counter Position

z sets the PSD/4's position counter to the value contained in the current encoder position.

Use *z* after a syringe overload error to resynchronize the PSD/4's actual position with its internally recorded position without having to go through the entire initialization sequence.



Re-initialization is recommended over the z command in order to ensure proper accuracy and precision.

Command Example	Description
zR	Set the PSD/4's position counter to the value contained in the current encoder position.

Table 8-6. Set counter position - Example

Ax - Absolute Position

A moves the syringe to absolute position *x*.

Parameter - absolute position *x* where $0 \leq x \leq 3000$ in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

Command Example	Description
A300	Moves syringe plunger to position 300.
A0	Moves syringe plunger to position 0.
A3000	Moves syringe plunger to position 3000;
Q	<i>Answer block</i> to <i>Query</i> shows pump busy (i.e., bit 5 is 0).

Table 8-7. Absolute position - Examples

Command Example	Description
IA1500OA0G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-8. Absolute position w/busy status - Example

ax - Absolute Position w/ready status

a moves the syringe to absolute position *x*.

Parameter - absolute position *x* where $0 \leq x \leq 3000$ in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

Pump status indicates Ready during the execution of this command.



Command Example	Description
a300	Moves syringe plunger to position 300.
a0	Moves syringe plunger to position 0.
a3000	Moves syringe plunger to position 3000;
Q	<i>Answer block</i> to <i>Query</i> shows pump ready (that is, bit 5 is 1).

Table 8-9. Absolute position - Examples

Command Example	Description
la1500Oa0G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-10. Absolute position w/ready status - Example

Px - Relative Pickup

P moves the syringe down *x* steps.

Parameter - number of steps *x* where $0 \leq x \leq 3000$ in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

Command	Description
IP1500OD1500G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-11. Relative pickup - Example

px - Relative Pickup w/ready status

p moves the syringe down *x* steps.

Parameter - number of steps *x* where $0 \leq x \leq 3000$ in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

Pump status indicates Ready during the execution of this command.

Command Example	Description
lp1500Od1500G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-12. Relative pickup w/ready status - Example



Dx - Relative Dispense

D moves the syringe up *x* steps.

Parameter - number of steps *x* where $0 \leq x \leq 3000$ in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

For example, the syringe is at position 3000. *D300* will move the syringe up 300 steps to an absolute position of 2700.

Command Example	Description
IP1500OD1500G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-13. Relative dispense - Example

dx - Relative Dispense w/ready status

d moves the syringe up *x* steps.

Parameter - number of steps *x* where $0 \leq x \leq 3000$ in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

Pump status indicates Ready during the execution of this command.

Command Example	Description
Ip1500Od1500G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-14. Relative dispense w/ready status - Example

Kx - Return Steps

K sets return steps to *x* steps.

Parameter - return steps *x* where $0 \leq x \leq 31$ in standard mode or $0 \leq x \leq 248$ in high resolution mode.



Any value given for *K* will reset to the default value upon syringe initialization.

Command Example	Description
K20R	Set return steps to 20.

Table 8-15. Execute command buffer from beginning - Example

**kx - Backoff Steps**

kx sets backoff steps to *x* steps.

Parameter - return steps *x* where $0 \leq x \leq 80$ in standard mode and $0 \leq x \leq 640$ in high resolution mode.

Command Example	Description
k50ZR	Initialize the syringe to the home position and set valve output position to the right side, move the syringe 50 backoff steps.

Table 8-16. Backoff Steps - Example



Valve commands

I - Move Valve to Input Position

I moves the valve to the input position set by the *Y* and *Z* commands.

See table 9-3 for input/output port location.

Command Example	Description
IA1500OA0G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-17. Move Valve to Input Position - Example

O - Move Valve to Output Position

O moves the valve to the output position set by the *Y* and *Z* commands.

Command Example	Description
IA1500OA0G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-18. Move Valve to Output Position - Example

B - Move Valve to Bypass (Throughput Position)

B connects the input and output positions, bypassing the syringe.

See Table 9-3, Valve input/output port locations, on page 9-14.

Command Example	Description
IA1500BR	Moves the valve to Input position, syringe to position 1500 then moves the valve to bypass.

Table 8-19. Move Valve to Bypass - Example

E - Move Valve to Extra Position

E moves the valve to the extra position (port) relative to the *Y* and *Z* commands.

Command Example	Description
IA1500ER	Moves the valve to Input position, syringe to position 1500 then moves the valve to the extra position.

Table 8-20. Move Valve to Extra Position - Example



Action commands



See “PSD/4 CAN Examples” on page 7-14, for additional CAN Action Command information.

g - Define a position in a Command String

g marks a position in a Command String that can be matched with *G* commands.

The *G* command is used with the *g* command to repeat commands within a Command String. *g* marks the start of the commands and is paired with *Gx* to mark the end of the commands and repeats them *x* number of times. Up to 10 pairs of *g* / *Gx* can be nested in a string.

Command Segment	Description
A0	Move syringe to position 0.
g	Outer loop start.
IP500	Move valve to input, move syringe down 500 steps.
OD500	Move valve to output, move syringe up 500 steps.
g	Inner loop start.
P150	Move syringe down 150 steps.
D150	Move syringe up 150 steps.
G10	Inner loop end, repeat ten times.
G5	Outer loop end, repeat five times.
R	Execute command.

Table 8-21. Repeat-sequence example for the command
A0gIP500OD500gP150D150G10G5R

Gx - Repeat Commands

G repeats a command in the command buffer *x* number of times.

Parameter - *x* where $1 \leq x \leq 65535$. For $x = 0$ and if *x* is omitted, the sequence is repeated until a terminate command is received at the PSD/4.

The *G* command allows the user to define the number of times a command in the Command String will be repeated. A *G* command without a matching *g* command repeats from the beginning of the command buffer.

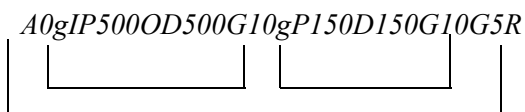


Command Example	Description
IA1500OA0G8R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-22. Repeat commands - Example

This is an example of using *g* and *G* to perform a nested loops. This example is a method to dispense a volume of liquid into a vessel 10 times, then go back and mix 10 times, and repeat the method 5 times:

A0gIP500OD500G10gPI50D150G10G5R



Mx - Delay

M performs a delay of *x* milliseconds.

Parameter - *x* where $5 \leq x \leq 30,000$ milliseconds.

Use the *M* command to pause the execution of the Command Buffer for a given amount of time.

Command Example	Description
M10000A3000R	PSD/4 waits 10 seconds after the Command String is sent, then moves syringe to absolute position 3000.

Table 8-23. Delay Example

Hx - Halt command execution

H halts execution of the commands in the command buffer. Execution of the command buffer can be resumed with an appropriate digital signal or with the execute command buffer command.

Parameter - *x* where

$x = 0$ - Waits for control command or either input 1 or input 2 to go from high to low

$x = 1$ - Waits for control command or input 1 to go from high to low

$x = 2$ - Waits for control command or input 2 to go from high to low

Unlike the *M* command, the *H* command is used to put an indefinite pause in a Command String or sequence. The operator can use an external device to trigger the Command String to resume.



The status of the digital input lines can be read using ?I3 and ?I4 commands as described in the Query Commands section of this manual.

Jx - Auxiliary Outputs

J sets the digital output lines.

Parameter - *x* where $0 \leq x \leq 7$ and is defined in table 8-24.

Use the *J* command to control the three digital outputs in the DB15 connector located on the back of the PSD/4.

PSD/4 Command	Output 3 (pin 15)	Output 2 (pin 14)	Output 1 (pin 13)
J0	0	0	0
J1	0	0	1
J2	0	1	0
J3	0	1	1
J4	1	0	0
J5	1	0	1
J6	1	1	0
J7	1	1	1

Table 8-24. Digital output control

Command Example	Description
J7R	Set digital outputs 1, 2, and 3 high.

Table 8-25. Auxiliary Output Example

sx - Store Command String

s stores the commands following the *s* command in the Command String in the specified EEPROM location.

Parameter - *x* where $0 \leq x \leq 14$ and *x* identifies the EEPROM location.

Use the *s* command to store the remaining commands in the Command String into the EEPROM. The Command String can then be executed by the controlling device, or upon power-up. (see Chapter 4, *Communications*)

Up to 15 Command Strings, numbered 0 through 14, can be loaded into the EEPROM. Each Command String contains up to 42 commands.

This is an example of how to store a Command String to execute a syringe movement of 1500 steps from an external controlling device.



Use H commands in the Command Strings to digitally control execution of the Command Strings stored in the EEPROM.

Command Example	Description
s2ZS4gIP1500OD1500H2GR	Store the following Command String in EEPROM location #2: Initialize the PSD/4 and set the syringe speed to 4. Start a loop. Move the valve to the Input position and move the syringe down 1500steps. Move the valve to the Output position and move the syringe up 1500 steps. Halt the command execution and waits for resume signal. Repeat from the start of the loop endlessly.

Table 8-26. Load command s2ZS4gIP1500OD1500H2GR into EEPROM

ex - Execute Command String in EEPROM Location

e executes the Command String stored in an EEPROM location.

Parameter - *x* where $0 \leq x \leq 14$. *x* identifies the EEPROM location.



Link Command Strings stored in the EEPROM by ending one Command String with an *ex* where *x* refers to the second Command String.

Command Example	Description
e2R	Moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

Table 8-27. Execute command string from EEPROM location - Example



Motor control commands

Nx - Standard/High Resolution Selection

Nx enables standard or high resolution mode.

Parameter - *x* is 0 or 1 where $x = 0$ for standard resolution mode and $x = 1$ for high resolution mode.

The PSD/4 uses 3000 steps/full stroke in standard resolution.

The PSD/4 uses 24000 steps/full stroke in the high resolution mode.

Command Example	Description
N1R	Enable high resolution mode.

Table 8-28. High resolution mode - Example



Lx - Set Acceleration

L sets the velocity ramp used by syringe moves to acceleration *x*.

Parameter - acceleration *x* where $0 \leq x \leq 20$.

Acceleration Code	Motor Steps per second/second
1	2500
2	5000
3	7500
4	10000
5	12500
6	15000
7	17500
8	20000
9	22500
10	25000
11	27500
12	30000
13	32500
14	35000
15	37500
16	40000
17	42500
18	45000
19	47500
20	50000

Table 8-29. Acceleration values *x* and the corresponding step rates

Command Example	Description
L2R	Set Acceleration to 5000 motor steps per second per second.

Table 8-30. Set Acceleration - Example

**vx - Set Start Velocity**

v sets the start velocity in motor steps/second the syringe begins its movement.

Parameter - *motor steps/second x* where $50 \leq x \leq 1000$.

Command Example	Description
v50R	Set start velocity to 50 motor steps per second per second

Table 8-31. Set Start Velocity - Example

Vx - Set Maximum Velocity

V sets the maximum velocity in motor steps/second.

Parameter - *motor steps/second x* where $5 \leq x \leq 5800$.

V is also an Async command. See “Async commands” on page 8-19.

Command Example	Description
V1000R	Set Maximum velocity to 1000 motor steps per second per second.

Table 8-32. Set Maximum Velocity - Example

Sx - Set Speed

S sets a predefined syringe maximum velocity.

Parameter - *pre-set syringe speed x* where $1 \leq x \leq 40$.

Speed Code	Motor steps per second	Maximum velocity in seconds per stroke
1	5600	1.2
2	5000	1.3
3	4400	1.4
4	3800	1.6
5	3200	1.9
6	2600	2.2
7	2200	2.6
8	2000	2.9
9	1800	3.3
10	1600	3.7
11	1400	4.3

Table 8-33. Speed codes with corresponding motor step rate and slew time



Speed Code	Motor steps per second	Maximum velocity in seconds per stroke
12	1200	5.0
13	1000	6.0
14	800	7.5
15	600	10.0
16	400	15.0
17	200	30.0
18	190	31.0
19	180	33.0
20	170	35.5
21	160	37.5
22	150	40.0
23	140	43.0
24	130	46.0
25	120	50.0
26	110	55.0
27	100	60.0
28	90	67.0
29	80	75.0
30	70	86.0
31	60	100.0
32	50	120.0
33	40	150.0
34	30	200.0
35	20	300.0
36	18	333.3
37	16	375.0
38	14	428.6
39	12	500.0
40	10	600.0

Table 8-33. Speed codes with corresponding motor step rate and slew time (Continued)



Command Example	Description
S11IA1500OA0G8R	Set syringe speed to 4.3 seconds per stroke, moves the valve to Input position, syringe to position 1500 then moves the valve to Output position and syringe plunger to position 0. The sequence is repeated 8 times.

*Table 8-34. Set Speed - Example***cx - Stop Velocity**

c sets the stop velocity in motor steps per second.

Parameter - motor steps/second x where $50 \leq x \leq 2700$.

Setting a stop velocity resets the cutoff steps to zero.

Command Example	Description
c500R	Set stop velocity to 500 motor steps per second.

*Table 8-35. Stop Velocity - Example***Cx - Increase Stop Velocity By Steps**

Cx increases the stop velocity by reducing the number of deceleration steps by the number of steps given.

Parameter - number of steps x where $0 \leq x \leq 25$.



It is recommended that cx be used for Stop Velocity control.

Command Example	Description
C10R	Increases stop velocity steps to 10.

Table 8-36. Increase Stop Velocity By Steps - Example



Async commands



See “PSD/4 CAN Examples” on page 7-14, for additional CAN Async Command information.

T - Terminate Command Buffer

T stops execution of the command buffer. It also aborts the command being executed, except for valve commands.

The *R* command may be used to resume the execution of the command buffer from the next unexecuted command.

T is used to terminate a command or Command Strings. *T* will not terminate a valve movement however it will terminate the Command String at the end of the valve move. Use *R* to resume the Command String or sequence.

T is an Async command.



If *T* was used to terminate a syringe move in mid stroke it may cause the motor to loose steps. The PSD/4 should be re-initialized after a syringe move is terminated or if an error occurs.

Command Example	Description
T	Stop execution of the command buffer.

Table 8-37. Terminate Command Buffer - Example

Vx - Set Maximum Velocity (on the fly speed change)

Vx is used to change the Maximum Velocity while the syringe is in motion.

Parameter - (on the fly speed change) motor steps/second *x* where $5 \leq x \leq 1024$.



V is an Async command when used for on the fly speed changes.



There are no ramps when changing the Maximum Velocity on the fly.

Command Example	Description
S40A3000R	Set syringe speed to 10 motor steps per second, move syringe to position 3000.
V1000R	Change syringe speed to 1000 steps per second.

Table 8-38. Set Maximum Velocity (on the fly speed change) - Example



Query commands



See "PSD/4 CAN query commands" on page 7-13, for CAN Query information.



A Control Command is not required to execute a Query Command.

F - Command Buffer Status

F reports the command buffer status

Command Buffer status	Return status code
Empty	0
Not Empty	1

Table 8-39. Report Buffer Status

& - Firmware Version

& reports the firmware revision in ASCII.

- Firmware Checksum

reports the firmware checksum.

Returned as 4 digit hexadecimal value.

Q - Pump Status

Q reports the pump status.

See 'Pump Status' definition.

? - Absolute Syringe Position

? reports the given position of the syringe.

Position is reported back where $0 \leq x \leq 3000$ steps in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

?1 - Start Velocity

?1 reports the start velocity in motor steps/second.

Start velocity is reported back in motor steps/second, $50 \leq x \leq 1000$.

?2 - Maximum Velocity

?2 reports the maximum velocity in motor steps/second.

Maximum velocity is reported back in motor steps/second, $5 \leq x \leq 6000$.

**?3 - Stop Velocity**

?3 reports the stop velocity in motor steps/sec.

Stop velocity is reported back in motor steps/second, $50 \leq x \leq 2700$.

?4 - Actual Position of Syringe

?4 reports the actual position of the syringe in steps based on encoder information.

Position is reported back where $0 \leq x \leq 3000$ steps in standard mode or $0 \leq x \leq 24000$ in high resolution mode.

?12 - Number of Return Steps

?12 reports the number of return steps.

Position is reported back where $0 \leq x \leq 31$ steps in standard mode or $0 \leq x \leq 248$ in high resolution mode.

?13 - Status of Auxiliary Input #1

?13 reports the Status of the Auxiliary Input #1.

0 - Auxiliary Input Low; 1 - Auxiliary Input High.

?14 - Status of Auxiliary Input #2

?14 reports the Status of the Auxiliary Input #2.

0 - Auxiliary Input Low; 1 - Auxiliary Input High.

?22 - Returns 255

?22 Returns 255.



Command summary

ASCII Command RS232/485	Parameters	Description	ASCII Command CAN
Control Commands			
R		Execute Command Buffer	R
X		Execute Command Buffer From Beginning	X
Initialization Commands			
Zx	$x = 0$ initializes at full plunger force; $x = 1$ initializes at half plunger force. $x = 10 - 40$, speed	Initialize PSD/4, assign output position to right side.	Zx
Yx	$x = 0$ initializes at full plunger force; $x = 1$ initializes at half plunger force. $x = 10 - 40$, speed	Initialize PSD/4, assign input position to left side.	Yx
Wx	$x = 0$ initializes at full plunger force; $x = 1$ initializes at half plunger force. $x = 10 - 40$, speed	Initialize PSD/4, configure for no valve.	Wx
Syringe Commands			
z		Reset syringe counter position	z
Ax	x where $0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Absolute movement to step position x	Ax
ax	x where $0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Absolute movement w/ ready status to position x step	ax
Px	x where $0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Pickup x steps	Px

Table 8-40. Command summary



ASCII Command RS232/485	Parameters	Description	ASCII Command CAN
px	x where $0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Pickup w/ready status x steps	px
Dx	x where $0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Dispense x steps	Dx
dx	x where $0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Dispense w/ready status x steps	dx
Kx	x where $0 \leq x \leq 31$ in standard resolution $0 \leq x \leq 248$ in high resolution	Set return steps to x steps	Kx
kx	x where $0 \leq x \leq 120$ in standard resolution $0 \leq x \leq 640$ in high resolution	Set syringe backoff steps to x steps	kx
Valve Commands			
I		Move valve to input position	I
O		Move valve to output position	O
B		Move valve to bypass position	B
E		Move valve to extra position	E
Action Commands			
g		Marks a position in a Command String that can be matched with G commands.	g
Gx	x where $1 \leq x \leq 65535$	Repeats the Command String x number of times. If $x = 0$ or x is omitted, the sequence is repeated until a terminate command is received.	Gx

Table 8-40. Command summary (Continued)



ASCII Command RS232/485	Parameters	Description	ASCII Command CAN
Mx	x where $5 \leq x \leq 30,000$	Executes a delay of x milliseconds.	Mx
Hx	$x = 0$ - Waits for R or either input 1 or input 2 to go from high to low $x = 1$ - Waits for R or input 1 to go from high to low $x = 2$ - Waits for R or input 2 to go from high to low	Halts execution of the commands in the command buffer, x defines resume parameter.	Hx
Jx	$0 \leq x \leq 7$	Sets digital output lines. See Table 8-24, Digital output control	Jx
sx	x where $0 \leq x \leq 14$	Stores all commands listed after the s command in the EEPROM location x .	sx
ex	x where $0 \leq x \leq 14$	Executes the Command String stored in an EEPROM location x .	ex
^x	x – is set to 255	(command ignored)	^x
Motor Control Commands			
Nx	x is 0 or 1 where $x = 0$ for standard resolution and $x = 1$ for high resolution	Syringe resolution	Nx
Lx	slope x where $0 \leq x \leq 20$	Set acceleration slope to x	Lx
vx	x where $50 \leq x \leq 1000$	Set start velocity to x motor steps/second	vx
Vx	x where $5 \leq x \leq 5800$	Set maximum velocity to x motor steps/second	Vx
Sx	x where $1 \leq x \leq 40$	Set syringe speed to preset speed x	Sx
cx	x where $50 \leq x \leq 2700$	Set stop velocity to x motor steps/second	cx
Cx	x where $0 \leq x \leq 25$	Increase stop velocity by x steps	Cx

Table 8-40. Command summary (Continued)



ASCII Command RS232/485	Parameters	Description	ASCII Command CAN
Async Commands			
T		Terminate Command Buffer	T
Vx	x where $5 \leq x \leq 1024$	Set maximum velocity for on the fly speed changes to x motor steps/second	Vx

Table 8-40. Command summary (Continued)



Query Commands			
RS232/485 Query	Response	Description	CAN Query
F	0 – Empty 1 – Not Empty	Report command buffer status	10
&	(string response)	Report firmware version	23
#	xxxx	Report firmware checksum in hexadecimal	
Q		Pump status	29
?	$0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Report absolute syringe position in steps	0 (zero)
?1	$50 \leq x \leq 1000$	Report start velocity in motor steps/second	6
?2	$5 \leq x \leq 6000$	Report maximum velocity in motor steps/second	4
?3	$50 \leq x \leq 2700$	Report cutoff velocity in motor steps/second	7
?4	$0 \leq x \leq 3000$ in standard resolution $0 \leq x \leq 24000$ in high resolution	Report Actual syringe position in steps	1
?12	$0 \leq x \leq 31$ in standard resolution $0 \leq x \leq 248$ in high resolution	Report number of return steps in steps	12
?13	0 - Auxiliary Input Low; 1 - Auxiliary Input High	Report status of auxiliary input 1	13
?14	0 - Auxiliary Input Low; 1 - Auxiliary Input High	Report status of auxiliary input 2	14
?22	Returns 255	Report 255	22

Table 8-41. Query Commands



Chapter 9

h Factor Command

- h factor command details
 - Enable/disable h factor commands
 - Syringe commands
 - Valve commands
 - Firmware commands
- Query commands
 - Syringe query commands
 - Valve query commands
 - System query commands
- Command summary
- Valve input/output port locations



The PSD/4 has a set of commands known as h Factor commands. These augment the capabilities of the PSD/4 by enabling the user to access expanded features such as multi-port valving and digital input/output controls.



In order to access the h Factor commands, they must first be enabled by sending h30001R to the PSD/4.



h factor command details

Enable/disable h factor commands

h30001 - Enable h Factor Commands and Queries

Enable h Factor Commands and Queries turns on the Hamilton Company extension commands.

h30000 - Disable h Factor Commands and Queries

Disable h Factor Commands and Queries turns off the Hamilton Company extension commands.

Syringe commands

h100xx - Initialize Syringe Only

Initialize Syringe initializes the syringe.

10000 + speed code

This command does not disable the valve like the *W* command.

h110xx - Set Syringe Mode

Set Syringe Mode configures the syringe.

xx is the sum of 11000 AND:

0 - standard resolution

1 - high resolution

0 - do not ignore overload

2 - ignore overload

0 - enable initialization sensor

4 - disable initialization sensor

0 - enable initialize

8 - disable initialize



Valve commands

h20000 - Initialize Valve

Initialize Valve initializes the valve.

h20001 - Enable Valve Movement

Enable Valve Movement enables the valve to be moved after valve movement was disabled.

h20002 - Disable Valve Movement

Disable Valve Movement makes the PSD/4 ignore all subsequent valve movement commands.

h2100x - Set Valve Type

Configure PSD/4 for specific valve type.

x is the sum of 21000 AND:

- 0 - 3-way 120 degree Y valve
- 1 - 4-way 90 degree T valve
- 2 - 3-way 90 degree distribution valve
- 3 - 8-way 45 degree valve
- 4 - 4-way 90 degree valve

h23001 - Move Valve to Input Position in Shortest Direction

Move Valve to Input Position in Shortest Direction moves the valve to the input position taking the shortest route in terms of degrees traveled.

h23002 - Move Valve to Output Position in Shortest Direction

Move Valve to Output Position in Shortest Direction moves the valve to the output position taking the shortest route in terms of degrees traveled.

h23003 - Move Valve to Wash Position in Shortest Direction

Move Valve to Wash Position in Shortest Direction moves the valve to the wash position taking the shortest route in terms of degrees traveled.

h23004 - Move Valve to Return Position in Shortest Direction

Move Valve to Return Position in Shortest Direction moves the valve to the return position taking the shortest route in terms of degrees traveled.

h23005 - Move Valve to Bypass Position in Shortest Direction

Move Valve to Bypass Position in Shortest Direction moves the valve to the bypass position taking the shortest route in terms of degrees traveled.

***h23006 - Move Valve to Extra Position in Shortest Direction***

Move Valve to Extra Position in Shortest Direction moves the valve to the extra position taking the shortest route in terms of degrees traveled.

h2400x - Move Valve in Clockwise Direction

Move Valve in Clockwise Direction moves the valve in a clockwise direction to one of eight positions.

$$1 \leq x \leq 8$$

h2500x - Move Valve in Counterclockwise Direction

Move Valve in Counterclockwise Direction moves the valve in a counterclockwise direction to one of eight positions.

$$1 \leq x \leq 8$$

h2600x - Move Valve in Shortest Direction

Move Valve in Shortest Direction moves the valve to position x in shortest direction in terms of degrees traveled.

$$1 \leq x \leq 8$$

h27xxx - Clockwise Angular Valve Move

Clockwise Angular Valve Move moves the valve to angle x in clockwise direction in 15° increments.

$$\text{Sum of } 27000 \text{ and } 0 \leq x \leq 345$$

h28xxx - Counterclockwise Angular Valve Move

Counterclockwise Angular Valve Move moves the valve to angle x in counterclockwise direction in 15° increments.

$$\text{Sum of } 28000 \text{ and } 0 \leq x \leq 345$$

h29xxx - Shortest Direct Angular Valve Move

Shortest Direct Angular Valve Move moves the valve to angle x in shortest direction in terms of degrees traveled.

$$\text{Sum of } 29000 \text{ and } 0 \leq x \leq 345$$

PSD/4 commands***h30003 - Reset PSD/4***

Reset PSD/4 resets the PSD/4 and sets power-up default values.



Query commands

Syringe query commands

?10000 - Syringe Status

Syringe Status queries the syringe. Response to query is decoded to determine the syringe status.

PSD/4 Response: *xx* where *xx* is decoded as the sum of:

- 0 - Syringe initialized
- 1 - Syringe not initialized

- 0 - No syringe stall or overload
- 6 - Syringe stall

- 0 - No initialization error
- 8 - Syringe initialization error

?10001 - Syringe Home Sensor Status

Parameters: none

PSD/4 Response: 1 - syringe in home region; 0 - syringe not in home region

?11000 - Syringe Mode

Syringe Mode queries the syringe. Response to query is decoded to determine the syringe mode.

PSD/4 Response: *xx* where *xx* is decoded as the sum of:

- 0 - standard mode set
- 1 - high-resolution mode set

- 0 - syringe overload not ignored
- 2 - syringe overload ignored

- 0 - enable initialization sensor
- 4 - disable initialization sensor

- 0 - enable initialize
- 8 - disable initialize



Valve query commands

?20000 - Valve Status

Valve Status queries the valve. Response to query is decoded to determine the valve status.

PSD/4 Response: *xx* where *xx* is decoded as the sum of:

- 0 - Valve initialized
- 1 - Valve not initialized

- 0 - No valve initialization error
- 2 - Valve initialization error

- 0 - No valve stall
- 4 - Valve stall

- 0 - Valve enabled
- 16 - Valve not enabled

- 0 - Valve is not busy
- 32 - Valve is busy

?21000 - Valve Type

Response to *Valve Type* is the valve type.

PSD/4 Response: *x* where $0 \leq x \leq 4$ and corresponds to:

- 0 - 3-way 120 degree Y valve
- 1 - 4-way 90 degree T valve
- 2 - 3-way 90 degree distribution valve
- 3 - 8-way 45 degree valve
- 4 - 4-way 90 degree valve



?23000 - Valve Logical Position

Response to *Valve Logical Position* is defined below in PSD/4 Response.

PSD/4 Response: x where x is defined as:

- 0 - not at logical position
- 1 - input
- 2 - output
- 3 - wash
- 4 - return
- 5 - bypass
- 6 - extra

?24000 - Valve Numerical Position

Response to *Valve Numerical Position* is defined below in PSD/4 Response.

PSD/4 Response: x where $0 \leq x \leq 8$ (0 corresponds to not a numerical position)

?25000 - Valve Angle

Response to *Valve Angle* is defined below in PSD/4 Response.

Parameters: none

PSD/4 Response: xxx where $0 \leq xxx \leq 345$

System query commands

?37000 - Last Digital Out Value

Reports the last digital output value

PSD/4 Response: x where $0 \leq x \leq 7$ corresponding to the last digital out values.



Command summary

Command	Parameters	Description
Enable/Disable h Factor Commands		
h30001		Enable h Factor Commands and Queries
h30000		Disable h Factor Command and Queries
Syringe Commands		
h100xx	pre-set syringe speed x where $0 \leq x \leq 40$ 10000 + ($0 \leq x \leq 40$)	Initialize Syringe at speed code x (Refer to Table 8-33, Speed codes with corresponding motor step rate and slew time, on page 8-16.)
h1100x	x where x is the sum of 11000 AND: 0 - standard resolution 1 - high resolution 0 - do not ignore overload 2 - ignore overload 0 - enable initialization sensor 4 - disable initialization sensor 0 - enable initialize 8 - disable initialize	Set Syringe Mode to x
Valve Commands		
h20000		Initialize Valve
h20001		Enable Valve Movement
h20002		Disable Valve Movement
h2100x	x = 0: 3-way 20 degree Y valve 1: 4-way 90 degree T valve 2: 3-way 90 degree valve 3: 8-way 45 degree valve 4: 4-way 90 degree valve	Set Valve Type to type x

Table 9-1. Command Summary



Command	Parameters	Description
h23001		Move Valve to Input Position in Shortest Direction
h28xxx	$22000 + 0 \leq xxx \leq 345$	Counterclockwise Angular Valve Move to angle x
h29xxx	$27000 + 0 \leq xxx \leq 345$	Shortest Direct Angular Valve Move to angle x
Firmware Commands		
h30003		Reset PSD/4

Table 9-1. Command Summary (Continued)

Query Commands		
Query	Response	Description
?10000	xx where xx is decoded as the sum of: 0 - Syringe initialized 1 - Syringe not initialized 0 - No syringe stall 6 - Syringe stall 0 - No initialization error 8 - Syringe initialization error	Syringe Status
?10001	1 - syringe in home region; 0 - syringe not in home region	Syringe Home Sensor Status

Table 9-2. Query Commands



Query Commands		
Query	Response	Description
?11000	xx where xx is decoded as the sum of: 0 - high-resolution mode off 1 - high-resolution mode on 0 - syringe overload not ignored 2 - syringe overload ignored 0 - enable initialization sensor 4 - disable initialization sensor 0 - enable initialize 8 - disable initialize	Syringe Mode
?20000	Response: xx where xx is decoded as the sum of: 0 - Valve initialized 1 - Valve not initialized 0 - No initialization error 2 - Syringe initialization error 0 - No valve stall 4 - Valve stall 0 - No valve positioning error 8 - Valve positioning error 0 - Valve enabled 16 - Valve not enabled 0 - Valve is not busy 32 - Valve is busy	Valve Status

Table 9-2. Query Commands (Continued)



Query Commands		
Query	Response	Description
?21000	x where $0 \leq x \leq 4$ and corresponds to: 0 - 3-way 120 degree Y valve 1 - 4-way 90 degree T valve 2 - 3-way 90 degree distribution valve 3 - 8-way 45 degree valve 4 - 4-way 90 degree valve	Valve Type
?23000	x where x is defined as: 0 - not at a logical position 1 - input 2 - output 3 - wash 4 - return 5 - bypass 6 -extra	Valve Logical Position
?24000	x where $0 \leq x \leq 8$ (0 corresponds to not at a numerical position)	Valve Numerical Position
?25000	xxx where $0 \leq xxx \leq 345$	Valve Angle
?37000	x where $0 \leq x \leq 7$ corresponding to the last digital out values	Last Digital Out Value

Table 9-2. Query Commands (Continued)



Valve input/output port locations



Table 9-3. Valve input/output port locations

	Input (Default Home)	Output	Bypass/ Extra
"T" Plastic			Bypass
"T" Ceramic			Bypass
"Y" Plastic			Bypass
3 - 5 Ceramic			Extra
4 - 2 Ceramic			Bypass
4 - 5 Ceramic			Extra
8 - 5 Ceramic			
6 - 5 Ceramic			



Appendix A Contacting Hamilton Company

In the United States and Canada

Hamilton Company • P.O. Box 10030 • Reno, Nevada 89520-0012

Phone

Technical Service: 800.648.5950 (8 a.m. to 4 p.m. PT)
Instrument Service:s 800.527.5269 (24 hours)

Fax: 775.856.7259

E-mail: tech@hamiltoncompany.com
sales@hamiltoncompany.com
service@hamiltoncompany.com

URL www.hamiltoncompany.com

Outside the U.S. and Canada

Phone: 775.858.3000

Fax: 775.856.7259

E-mail: tech@hamiltoncompany.com
sales@hamiltoncompany.com or
service@hamiltoncompany.com

URL: www.hamiltoncompany.com

In Switzerland

Hamilton Bonaduz AG • Ch-7402 • P.O. Box 26 • Bonaduz, Switzerland

Phone: 41.81.660.60.60

Fax: 41.81.660.60.70

E-mail: itech@bonaduz.hamilton.ch
hamopd@bonaduz.hamilton.ch

URL: www.hamiltoncompany.com





MICROLAB[®] PSD/4

PSD/4 User Manual
Part No. 8892-01 (Rev. B)